TIMETABLE MODULE (CORE ENGINE)

SmartCampus School Management System
Technical Specification & Implementation Guide — v1.0 • March 2026
Confidential • Internal Documentation

Audience: School Administrators • Teachers • IT Team • Students • Parents
Related Modules: Academics • Curriculum • Student Management • Staff Management • Notifications

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1. MODULE OVERVIEW

1.1 Purpose

The Timetable Module provides a central platform for creating, managing, and monitoring class schedules across grades, teachers, and classrooms. It supports efficient lesson delivery and staff coordination by offering:

- Weekly and term-wide timetable views
- Class- and teacher-specific schedule filtering
- Integration with curriculum coverage and academic modules
- Alerts and notifications for schedule changes
- Reporting and analytics for resource utilization

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2. CORE OBJECTIVES

1. Enable administrators to create and manage school timetables efficiently
2. Provide personalized timetable views for teachers and students
3. Detect conflicts and overlaps automatically
4. Link timetable changes to notifications and parent/student portals
5. Provide analytics on teacher workload, classroom utilization, and schedule efficiency
6. Support mobile-first and accessibility-compliant design
7. Allow scalable scheduling for multi-campus operations

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3. REAL-WORLD SYSTEM FLOW

3.1 Lifecycle Flow

Term Timetable Created → Subjects & Teachers Assigned → Lessons Scheduled → Conflicts Detected → Staff & Student Views Updated → Changes Notified → Analytics Generated

3.2 Daily Operational Flow

Administrator/teacher opens timetable
↓
Add or adjust slots (manual or template-based)
↓
System checks for conflicts (teacher/room overlaps)
↓
Timetable syncs to all authorized users (teachers/students)
↓
Notifications sent for changes
↓
Usage analytics updated in dashboard

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4. MODULE ARCHITECTURE

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Timetable Module
│
├── Weekly/Term Grid Views
├── Class & Teacher Filtering
├── Slot Management (Add/Edit/Delete)
├── Conflict Detection Engine
├── Notifications Integration
├── Analytics & Reports
├── Student & Teacher Portal Integration
├── Mobile & Offline Support
├── Accessibility & Localization
├── Security & Role-Based Access
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5. TIMETABLE GRID

5.1 Features

- Visual weekly/term timetable for each class and teacher
- Filter by class, teacher, or room
- Print-friendly view for administration
- Quick link to curriculum topics or schemes of work

5.2 Data Model

TimetableSlot

- slot_id
- class_id
- teacher_id
- subject_id
- room_id
- start_time
- end_time
- day_of_week
- status (scheduled/cancelled/rescheduled)

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6. SLOT MANAGEMENT

6.1 Features

- Add, edit, or delete timetable slots
- Bulk slot creation and template-based scheduling
- Drag-and-drop functionality (planned upgrade)
- Conflict detection for double-booked teachers or rooms

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7. CONFLICT DETECTION & AUTOMATION

7.1 Features

- Identify overlapping lessons for teachers or classrooms
- Suggest alternative times automatically
- Integrate with Academic & Curriculum modules to ensure coverage compliance

7.2 Rules

IF teacher double-booked OR room occupied
→ Trigger alert
→ Suggest alternative slot
→ Notify affected users

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8. NOTIFICATIONS

8.1 Features

- Push notifications or emails for timetable changes
- Alerts for upcoming classes, cancellations, or rescheduled lessons
- Integration with parent and student portals

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9. ANALYTICS & REPORTING

9.1 Features

- Teacher workload by hours/subjects
- Classroom utilization rates
- Schedule efficiency and gap detection
- Historical reporting for term reviews

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10. STUDENT & TEACHER PORTAL INTEGRATION

10.1 Features

- Personalized timetable view for each student and teacher
- Sync with mobile app for on-the-go access
- Offline viewing of schedules with auto-sync

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11. ACCESSIBILITY & LOCALIZATION

11.1 Features

- Multilingual support for interface
- Compliance with accessibility standards (screen readers, contrast, navigation)
- Mobile-friendly responsive design

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12. SECURITY & ROLE-BASED ACCESS

Role	Access Scope
Admin	Full timetable creation, edits, reports
Teacher	Assigned class timetable view, edit own slots
Parent	Child's timetable view only
Student	Own class timetable view only

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13. MOBILE-FIRST & OFFLINE DESIGN

13.1 Requirements

- Fully responsive timetable grid on tablets and smartphones
- Offline caching of schedules for low-connectivity environments
- Automatic sync upon reconnect

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14. SCALABILITY

- Multi-campus support with thousands of students
- Quick filtering for large datasets
- Extendable for additional timetable types (extracurricular, exams, interventions)